UNITED STATES OF AMERICA
GENERAL SERVICES ADMINISTRATION

FYS

Federal Supply Service
Washington, DC 20406

CLARIFICATION - NOTICE

TO

Amendment - 1 of Fed. Spec. KKK-A-1822

Ambulance

Due to an oversight in paragraph 3.14.6 on siren's, some detailed data may be desirable. Although reputable ambulance producers will provide quality siren equipment, it is suggested the following paragraph should be substituted in your procurement documents, or changed to read accordingly

3.14.6 Siren - public address system. The combined electronic siren (with warning light(s) permitted) and public address system furnished shall include upgrading features; radio amplification capability, complete with speaker (s) (see 3.14.7) and noise cancelling microphone. The siren control functions and related characteristic sound patterns shall consist of manual and automatic wail, yelp, and "Hi-Lo" sound when applicable or specified. Plug-in connections and connecting cables, foot switch and all hardware necessary for normal siren/speaker(s) operation shall be furnished. The siren unit shall have an amplifier capable of producing continuous warning tones with a power output of not less than 80 watts RMS with a one-speaker load and not less than 130 watts RMS with a two-speaker load when the amplifier is supplied with 13.6 volts dc input. The speaker(s) furnished shall have a minimum 100 watt RMS rating. In voice operation, the power output shall be a nominal 35 watts, with not more than 10 percent voice distortion from 300 to 3000 Hz. The public address amplifier shall be independent of the two-way (mobile) radio except that a common microphone and control housing group may be employed. The siren assembly (including parts installed in the passenger space of the emergency vehicle) shall pass the radio noise interference tests specified in SAE Standard J551B.

Note: New data underlined.

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8-26-75

FEDERAL SPECIFICATION

AMBULANCE

EMERGENCY MEDICAL CARE VEHICLE

This amendment, which forms a part of Federal Specification KKK-A-1822, dated January 2, 1974, was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

PAGE 1

Paragraph 1.2.1: Type II, class 2 delete: "(Not yet available)," Substitute: "Four wheel driven (4x4)."

PAGE 2

Paragraph 2: Applicable documents, add to bottom of page:

Federal Communication Commission (FCC):

Medical Communication Service
Title 47, CFR, Part 89 - Public Safety Radio Services

PAGE 3

Paragraph 2.2: Delete: "J551a" and substitute:

J551B - Measurement of Electromagnetic Radiation from a Motor Vehicle or other Internal-Combustion-Powered Device (excluding aircraft) (20-1,000 MH_z).

PAGE 5

Paragraph 3.1.3: Type II ambulance, after "class 1" on the first line add: "or 2".

PAGE 9

Paragraph 3.6.4.5: Delete entirely and substitute:

3.6.4.5 Cooling system. The heaviest-duty cooling system components, and capacity, offered by the chassis manufacturer shall be furnished, and a coolant recovery system shall be provided. The cooling system and related engine components shall meet the following test requirements:

a. With transmission in neutral or park, and full air conditioning load applied, the engine shall normally idle for 40 minutes in ambient temperatures of 95° F. and;

b. The vehicle burdened in accordance with 3.4.1, the ambulance shall travel not less than 5 miles at 30 m.p.h. ± 1 m.p.h. on a road of at least 7 percent (± 2 percent) grade to altitudes up to 8,000 feet. The cooling system shall maintain normal operation without any loss of coolant or overheating of the engine and related components.

PAGE 10

Paragraph 3.6.5.7: Brake systems, after the last sentence add: "Exception to front disc type brakes is permitted for class 2 vehicles when unavailable on the chassis offered."

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Paragraph 3.7.10: Delete entirely and substitute:

3.7.10 Electronic and radio interference suppression. The ambulance (the final stage manufacturer) shall have sufficient electromagnetic radiation suppression to prevent emitting interference to any specified electronic, biophysical monitoring, and radio(s) equipment furnished.

Paragraph 3.8.2: Delete entirely and substitute:

3.8.2 Warning lights. Unless otherwise specified (see 6.2), a roof top-mounted 360° rotating dome beacon lamp containing four sealed beam bulbs, each clear lamp having a minimum 35,000 candle power, with alternating lens colors, two white and two blue, lights shall be provided and located near the center of vehicle or body. The upper body, or the roof corners shall also be provided with alternate flashing pattern blue color lights. These corner warning lights, dome or tunnel style showing two on the front facing front, and two on the rear facing rearward, shall be the seal beam type having a minimum of 500 candle power output each. In addition at least two regular blue flashing warning lights, shall be located in, on, behind, or in front of the vehicles radiator grill. Warning lights shall comply with SAE J845 and J595b. Warning light mountings shall be provided with reinforcements at point of attachment to vehicle. Warning lamps shall not be obstructed by open doors.

PAGE 13

Paragraph 3.8.6: Delete entirely and substitute:

3.8.6 Patient compartment illumination. Normal white illumination in all patient areas shall be not less than 15 foot candle intensity measured all along the center line of the clear floor with doors open and all ambient light obilterated. Patient compartment lighting shall also be capable in combination, or by separate means to provide at least 40 foot candle intensity, when measured at any point on top of the cots. A reduced lighting level controlled by the technician shall also be provided by switch(s), electronically, or with a fire-proofed rheostat to control compartment lighting, or by a second system of low intensity lights. Blue light(s) or lens shall not be used in the patient compartment. The patient compartment (normal) light(s) and step-well lights shall be automatically activated when opening the entrance doors, otherwise controlled by drivers master switch. Fixtures (interior) shall be near flush mounted protruding not more than a nominal 1.5 inches.

Paragraph 3.8.7.2: Delete the following:

"Switches shall be located at the rear doors, and be operable from ground level reach to control the following:

Exterior area flood lights (separate switches)".

PAGE 14

Paragraph 3.9.2 Cab-body provisions, after item "(e)" Ammeter add: "or voltmeter".

Paragraph 3.9.5: Delete entirely and substitute:

3.9.5 Outside rear view mirrors. Unless otherwise specified (see 3.22), dual, below-eye level, or western style outside rear view mirrors, with a combination or separate wide angle mirror system shall be furnished. Mirrors shall have the largest usable reflective area practical for the designed ambulance, totalling at least 125 square inches per vehicle. All mirror heads shall be independently adjustable. Hardware and mirror heads shall be of polished metal.

PAGE 15

Paragraph 3.10.3: On the second line change the word "headrest" to "backrest".

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PAGE 17

Paragraph 3.10.11: Delete entirely and substitute:

3.10.11 Floor coverings and color. Floor covering shall harmonize with the interior color and decor of the patient's compartment. The floor covering shall be seamless, one piece no waxing type linoleum or vinyl not less than 1/16 inch thick, permanently applied, and shall extend the full length and width of the compartment. Covering joints at the side walls (where side panels and covering meet) shall be sealed and bordered with rustproof, corrosion-resistant metal cove moulding.

Paragraph 3.10.13 Wheelhousings. Add to the first line after the word "include", "metal or fiberglass".

PAGE 18

Paragraph 3.10.16: Delete entirely and substitute:

3.10.16 Insulation. The entire body, sides, ends, and roof of the patient's compartment shall be insulated to enhance the environmental criteria specified (3.4.2 and 3.13) minimizing conduction of heat, cold, and external noise entering the vehicle interior. The insulation shall be vermin-and mildew-proof, fire-retardant, non-hygroscopic, and a non-settling type. Undercoated doors and floor will be considered insulated.

Paragraph 3.11.1: Second line after the word "minimum", add "volume".

PAGE 19

Paragraph 3.11.2: Fourth line, after "Space for" add "at least".

Under: Approximate area in cubic feet change "4" to "10".

Paragraph 3.11.4: Eighth line: after "not less than", change "3 inches thick" to read, "2.5 inches thick or more than 3.5 inches thick". Next to the last line, after "furnished with 3" insert "sets of" safety belts.

PAGE 21

Paragraph 3.12.2.1: Sixth and tenth lines down and in front of, "100 percent oxygen" insert the word, "practically".

PAGE 22

Paragraph 3.13.6: Delete entirely and substitute:

3.13.6 Ventilation criteria. Ventilation system(s) of the driver and patient compartments shall provide a complete change of ambient air within each compartment every two minutes. Ventilation shall be separately controlled within each compartment. Fresh air intakes shall be located in the most practical contaminant-free air space on the vehicle. The patient compartment shall be ventilated by the air delivery system of the environmental equipment (heater-air conditioner) or by separate system(s); such as power intake, exhaust ventilator(s), and static roof vent(s).

Paragraph 3.13.7: Delete entirely and substitute:

3.13.7 Environmental controls. Adjustable, manual, or thermostatically operative controls shall permit heating and/or air conditioning and ventilating in either compartment, without affecting the other compartment or simultaneously heating one compartment and air conditioning the other. Switches and controls shall be located in panels and be readily available and identified for function and operating position (see 3.8.7 thru 3.8.7.2). Switches and control components shall exceed in capacity the amperage and resistance requirements of the motors. Blower or fan motors shall be at least three speed design. Shut-off valve(s) for the hot water heating system of the patient compartment shall be provided to allow servicing and shut off. Air systems shall have adjustable louvers to direct the flow of air.

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Delete entire page and substitute new page.

3.14 Communications.

3.14.1 Communication equipment. Ambulances will be equipped with two-way radio(s) (mobile and walkie-talkie), intercom, public address, electronic siren, and telemetry system(s). These systems when specified herein or purchaser furnished shall be installed in ready to-use mode. Switches and controls shall be panel mounted in accordance to 3.8.7 with switch positions identified. Communication equipment should be consistent with the overall medical radio plan for the service area of ambulance operation and in accordance with the regulations of the Federal Communication Commission (referenced in section 2).

3.14.2 Radio (mobile) provisions. All ambulances shall be provided with sufficient ventilated space for a modern two-way radio, convenience features (antenna openings, ground plate, terminal wiring and panels, etc.), required for the installation, and servicing of mobile radios. Unless otherwise specified (see 6.2), the consignee will locally install mobile radio unit(s) or shall provide any additional data, equipment, and a coordinator to insure compatibility of specified systems.

3.14.3 Antenna, cable and mountings. Antenna when furnished shall meet the requirements of EIA Standard RS329 Part II. The coaxial lead-in wire furnished shall be of the impedance, loss, and power handling capability specified by the mobile radio manufacturers. The contractor shall provide each ambulance a ground plate and opening, and coaxial lead-in wire from the ventilated radio storage compartment to the antenna port.

3.14.4 Two-way (mobile) radio equipment. When specified (see 6.2), a solid state (no tubes) mobile two-way radio shall be furnished, with all normal accessories. This radio shall meet the requirements of EIA Standards RS152B (transmitters), RS204A (receivers), RS329 Part II (mobile antennas), and, if applicable RS220 (continuous tone coded squelch systems) and RS374 (selective signalling). The transmitter power output shall be *(as specified) watts minimum. Microphones shall be provided for transmitting at both the driver's position and in the patient's compartment. Receiver audio power output shall be *(as specified) watts minimum. Selectable speaker outputs, singularly and in combination shall be provided at the driver's position, patient's compartment, and through the public address system. Radio interference suppression see 3.7.10.
*Power to be specified (see 6.2), consistent with local emergency communication requirements.

3.14.5 Intercom system. A solid state inter communication facility shall be provided between the driver and patient compartments. The driver shall have the capability to place the patient compartment in stand-by 'talk' mode and to control any necessary talk/listen switching. The intercom system shall be independent of the two-way (mobile) radio equipment or can be supplied in conjunction with the radio equipment providing that use of the intercom does not prevent the receipt of radio messages or interfere with the transmitting capability from the patient compartment.

3.14.6 Siren - public address system. The combined electronic siren (with warning light(s) permitted) and public address system furnished shall include upgrading features; radio amplification capability, complete with speaker(s) (see 3.14.7) and noise cancelling microphone. The siren control functions and related characteristic sound patterns shall consist of manual and automatic wail, yelp, and "hi-Lo" sound when applicable or specified. Plug-in connections and connecting cables, foot switch and all hardware necessary for normal siren/speaker(s) operation shall be furnished. The siren unit shall be capable of powering a nominal 100 watts speaker with 13.5 volts d.c. power input. In voice operation, the power output shall be a nominal 35 watts, with not more than 10 percent voice distortion from 300 to 3000 Hz. The public address amplifier shall be independent of the two-way (mobile) radio except that a common microphone and control housing group may be employed. The siren assembly (including parts installed in the passenger space of the emergency vehicle) shall pass the radio noise interference tests specified in SAE Standard J551B.

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3.14.7 Speakers. One or two (siren/public address/radio) speakers as required shall be installed outside on the vehicle. Wherever practical, metals used in manufacturing shall be either corrosion-resistant (polished metal finish) or treated to resist corrosion (painted white). The use of dissimilar metals, especially brass, copper or steel in intimate contact with aluminum or aluminum alloy shall be avoided wherever practical. Design and construction shall ensure that no parts will become loose and shall withstand the strains, jars, vibrations, shocks and other similar conditions incident to service operations. Installation shall be on reinforced mounting to prevent drumming and vibration. The siren/speaker manufacturer shall provide detailed instructions for installation. The speaker(s) shall be mounted facing forward, located near the longitudinal centerline and front section of the emergency vehicle. Speaker location shall permit servicing of the vehicle without removing the speakers or interfering with the intent or vision of the warning lights. If dual speakers are employed, the manufacturer shall provide the following information:

a. The angle between the acoustic axis of each speaker and the longitudinal centerline of the vehicle.

b. Electrical connections specifying whether the speakers are connected in phase or out of phase.

c. Directionality plots or directionality response data for each suggested mounting configuration when applicable.

Electronic sirens, amplifier boxes, drivers and speakers shall be permanently marked with the manufacturer's name, initial or lettered trademark and the model designation in letters and numerals nominally 1/8 inch in height. The power (wattage) rating of the speaker driver specified by the manufacturer of the driver shall be clearly marked on the driver. The normal operation of the siren shall be such that the power delivered to the driver shall not exceed the power rating of the driver.

PAGE 24

Paragraph 3.15.2: After item e., second line delete "with padded ankle-hitch or skin-traction device, with either a built-in turnbuckle device or a supplemental 'Spanish windlass' device".

Paragraph 3.15.3: Delete entirely and substitute:

3.15.3 Optional equipment. When specified (see 6.2), in addition, the ambulance shall be equipped with, but not limited to the following items:

1. Battery charger; with electric "T" polarity plug-in outlet to 110 volt receptacle.
2. Tachograph; 90 mph, with 24-hour chart, plus light and siren recording.
3. Intercom; two-way visible intercommunication system between driver and technician with amber, red and green lights.
4. Mud-guards; to protect lower front-end of body (for type I).
5. Backup alert; audible warning device activates when vehicle is shifted into reverse.
6. 110 volt, ac electric power systems (see 3.7.8)
   a. 1,000 watts, for bio-medical and lighting, etc.
   b. 3,000 watts, as for a above and rescue tools.
7. Drug compartment; security compartment, minimum 8" x 8" x 12" long; with hidden hinge, door with key lock in patient compartment.
8. Power winch, electric; front center mounted, minimum 3 ton capacity, with wire rope, chain, hook, and roller guides.
9. Rear door medicine compartment; recessed in lower interior panel of door, or attached to rear door(s).
10. High-intensity cot light; with flexible or adjustable shaft located near head of primary cot.
11. Patient fan; located near primary cot, minimum 250 CFM rating.
12. Rear step stowage compartment, (types I and III).
13. Partition; with access door between compartments for types II and III (see 3.10.2).
14. Window; clear, right side patient compartment, minimum size 430 square inch area (12" x 36" long).
15. Drapes; fire-retardant cloth drapes for each window of patient compartment except partition; installed on polished metal drapery rods.

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16. Power plant heaters; minimum -30 degree F. startability (see 3.6.3.2.1, severe cold area only).
17. High altitude operations, engine emissions adjustments and controls (see 3.6.4.3).
18. Extrication equipment, as specified (see 3.11.2).

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Paragraph 3.15.4: Delete item after "M1" and substitute the following:

M1 - Rail system; for attaching various medical devices to the rail with quick release guides; rail length shall be 60 to 90 inches, mounted near the primary cot, example: Modura (Fairfield) Rail System or equal.

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Paragraph 3.16.3.1: Delete entirely and substitute:

3.16.3.1 Color standards and tolerances. The exterior surface including the wheels shall be manufacturer's standard gloss white. The ambulance colors orange (stripe or band) and blue (markings) shall be the same as those specified as Safety Orange and Safety Blue in American National Standard Z53.1-1974, Safety Color Code for Marking Physical Hazards. They shall comply with the following tolerances expressed in terms of Munsel hue, value (lightness), and chroma (saturation) as follows:

Ambulance Orange, 5.OYR 6.0/15 with tolerances in hue, value, and chroma of: Hue+, 6.25YR; Hue-, 3.75YR; Value+, 6.5; Value-, 5.5; Chroma+, unlimited; and Chroma-, 13.

Ambulance Blue, 2.5PB 3.5/10 with tolerances in hue, value, and chroma of: Hue+, 4.5PB; Hue-, 10.OB; Value+, 4.0; Value-, 3.0; Chroma+, unlimited; and Chroma-, 8.

(Color tolerance charts containing the color standards and tolerances for Ambulance Orange and Ambulance Blue will be available from the National Highway Traffic Safety Administration (NHTSA, Office of Hazardous Materials), 400 7th Street, Washington, DC 20590.)

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Paragraph 3.16.4: After 3rd line "(color a)" add: " , and white (color i)".

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After paragraph 4.1 add:

4.1.1 Certifications and qualified contractor/manufacturers. The supplier is obligated to prove that the vehicle, ambulance components and equipment offered/provided meet or exceed the requirements of the contract. When specified (see 6.2), the supplier/contractor shall, certify prior to bid opening that he possesses the following preliminary certified verification data to support items: a, b, c, i, k, l, m, n, and q under tests in 4.4.2, qualifying the contractors/manufacturer currently offered ambulance/equipment. Two copies of each, the letter of certification, manufacturer's specification data, pictures, and brochures of the tested/offered ambulance shall be forwarded to the procuring activity with the bid for preliminary review.

Paragraph 4.4.1: Delete entirely and substitute:

4.4.1 Test criteria and certifications. The ambulance shall be prepared for operation in accordance with manufacturer's recommendations and 3.21 manned and loaded to simulate maximum payload conforming to type, class and plan specified. Road test course shall be approved by the procuring activity, or purchaser representative. Substitution of new vehicle or replacement of major component(s) may require complete retest at the discretion of the purchasers representatives. To avoid duplication, certification(s) may be acceptable in lieu of actual verification during inspection, providing supporting verifying data is on file for examination. The certifications will remain valid as long as they are applicable to the vehicle model, component and equipment specified, so offered and furnished. Certifications that appear on the vehicle need not be resubmitted (i.e., Department of Transportation, DOT; Environmental Protection Agency, EPA; etc.) Certifications with verifying test data for Federal procurements shall be forwarded to the Office of Standards and Quality Control (FMQ), General Services Administration, Federal Supply Service, Washington, DC 20406 and shall also be provided on request to other authorized agents.

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Paragraph 6.2: Delete entirely and substitute:

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement document:

(a) Title, number, and date of this specification, and amendment number if any.

(b) Quantity, type, and class ambulance required (see 1.2.1).

(c) Patient compartment floor plan (plan A is standard); specify if plan B is desired (see 3.1.5).

(d) State if California Emission Controls and test required (see 3.6.4.3).

(e) Manual four speed transmission, for class 2 (4 x 4) and transfer case, automatic transmission is standard, detail other powertrain requirements. Check supplier's option and availability, (see 3.6.5.2.1 and 3.6.5.3).

(f) Axle ratios, state if operating in mountainous terrain, (do not specify ratios) (see 3.6.5.6).

(g) Wheels, specify if dual rear wheels are specifically desired for types I or III vehicle (see 3.6.7).

(h) Tires, specify where snow tires instead of regular tires desired (see 3.6.8 and 3.6.9).

(i) Warning lights, (blue and white colored lights are standard), specify if other colors are required by state or local regulations (see 3.8.2).

(j) Communication equipment, two-way radio and siren (local radio installations are recommended) however, if contractor is to provide the radio communication equipment specify, cite brand(s) acceptable, power output and frequencies required. Give complete details for installation(s) and furnish a coordinator's name, address, and telephone number (see 3.14 to 3.14.7).

(k) Additional systems, equipment, accessories, and supplies (see 3.15 thru 3.15.4), cite additional parameters where and when necessary.

1. Select optional (vehicle) equipment desired, 1 thru 18 (see 3.15.3).
2. Select medical (ambulance) equipment desired, M1 thru M14 (see 3.15.4).

(l) Rustproofing, if required, undercoating is standard (see 3.18).

(m) Special requirements, state additions, changes, or deletions of equipment items from specification (see 3.22).

(n) Quality assurance provisions, specify if bidders certification and specifications qualifying offered ambulance is required (see 4.1.1).

(It is recommended that all "Ambulances" procured under this specification be inspected at the supplier's plant for quality assurance, in accordance with at least (c) of 4.2 Quality conformance inspection (see 4.9).)

1. State which classification(s) of inspection desired (see 4.2).
2. First production ambulance inspection, recommended for column procurement only.

(o) Preparation for delivery, other than specified (see 5.1). State mode of delivery preferred.

(p) Procurement requirements (see 6.3).

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